SAGE: Spectral Artifact-Guided Decision-Based Evasion of Diffusion-Generated Image Detectors

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Abstract

Diffusion-generated image detectors may rely on statistical traces that can be modified without changing the intended image content. Evaluating this vulnerability under a label-only interface requires attacks that balance detector evasion, query expenditure, and image distortion. We propose SAGE, a spectral artifact-guided decision-based evasion framework. Spectral Artifact Profiling estimates a signed difference between the average power spectra of independent real and generated development images. Adaptive Spectral Alignment then selects image-specific correction strengths over radial frequency bands by balancing spectral-distribution alignment with a reconstruction-based perceptual penalty. The corrected magnitude is combined with the input phase to produce a detector-independent candidate. If that candidate does not satisfy both the evasion and distortion requirements, a verified real-classified starting point is refined using boundary search with spectrally weighted random directions. All attack-stage detector calls are counted within one query budget, and success requires a real prediction within a predefined distortion constraint. The proposed evaluation compares complete attack pipelines and controlled refinement variants, including cross-generator transfer, component ablations, and image-quality assessment. Experimental findings remain pending; the present draft defines the method and evaluation protocol rather than establishing superiority over existing attacks.

Keywords: diffusion-generated images; image forensics; decision-based attack; spectral prior; black-box robustness

1 Introduction

Generative image models support creative and analytical applications but also complicate the assessment of image authenticity. Forensic detectors exploit learned representations, frequency statistics, or reconstruction errors to distinguish generated images from photographs. Examples include frequency-based recognition [1], pretrained visual representations [2], diffusion reconstruction error [3], and autoencoder reconstruction error [4]. Their accuracy under ordinary evaluation conditions does not establish robustness to intentional manipulation.

Detector-independent anti-forensic transformations seek to weaken generation traces without repeatedly querying a target detector. Examples include GANprintR, FakePolisher, Deep Dithering, and DeepNotch [5, 6, 7, 8]. Such transformations optimize indirect objectives; improved distributional similarity need not imply a changed detector decision. Conversely, decision-based attacks use output labels to adapt to the actual target boundary [9, 10, 11]. Their search can be expensive when initialization is distant from the source image or exploration directions are poorly matched to the task.

SAGE combines these two mechanisms. A development-set spectral discrepancy provides a detector-independent correction direction and a frequency weighting for online exploration. The hypothesis is that this shared prior can improve the evasion–distortion–query trade-off. The discrepancy is not assumed to identify detector gradients, and successful spectral alignment is not assumed to guarantee evasion. Both propositions require empirical testing.

The framework has three components: Spectral Artifact Profiling (SAP), Adaptive Spectral Alignment (ASA), and Spectral-Biased Query Refinement (SBQR). SAP estimates real–generated power-spectrum

differences. ASA adapts their correction strength to each image while penalizing reconstruction distortion. SBQR uses the correction-derived weights to allocate random perturbation energy across frequencies and refines a real-classified point toward the original generated image using binary feedback.

The methodological contributions are (i) a shared, development-set spectral prior for image transformation and label-only refinement; (ii) band-wise spectral alignment coupled through a full-image perceptual penalty; and (iii) an evaluation formulation that requires evasion within a distortion constraint and counts initialization together with refinement queries. Whether these components improve performance is assessed through the proposed comparisons and ablations. No claim of empirical superiority or comprehensive novelty is made before experiments and a broader literature audit are completed.

2 Related Work

2.1 Generated-Image Detection and Counter-Forensics

Frequency artifacts have been used for generated-image recognition [1]. Generalization across generators motivates detectors based on pretrained visual representations [2], while DIRE and AEROBLADE exploit reconstruction behavior [3, 4]. Spectral learning provides another approach to generated-image detection [12]. These different forensic mechanisms motivate evaluation across detector families rather than assuming that every detector uses a global spectral signature.

GANprintR [5], FakePolisher [6], Deep Dithering [7], and DeepNotch [8] transform synthetic images to weaken forensic traces. Diffusion-based counter-forensics also investigates generative reconstruction as a means of suppressing manipulation evidence [13]. These studies address different image domains and training requirements. Their applicability to diffusion-generated images must be checked before treating them as directly comparable baselines.

2.2 Decision-Based Attacks and Biased Exploration

Boundary Attack searches from a point already assigned to the desired class [9]. HopSkipJumpAttack estimates boundary directions using label feedback [10], and QEBA uses reduced-dimensional boundary estimation [11]. SurFree and Triangle Attack provide alternative geometric searches [14, 15]. Low-frequency perturbations [16] and biased Boundary Attack [17] show how image priors can shape black-box exploration.

SAGE builds on biased boundary exploration rather than proposing an entirely new boundary-search geometry. Its intended distinction is the reuse of a real–generated spectral discrepancy in both initialization and frequency-weighted refinement. Unlike an explicit low-dimensional restriction, positive weights retain full-dimensional support. Matched-initialization comparisons against isotropic and fixed low-frequency sampling test whether any gain comes from the learned spectral allocation rather than a better starting point or generic low-frequency bias.

3 Methodology

This section consolidates the adaptive correction and online boundary refinement of the updated methodology. SAP is computed offline once per profiling condition. ASA and SBQR operate on each test image.

3.1 Threat Model and Constrained Objective

Let $x_o \in [0,1]^{H\times W\times 3}$ be a generated image and let $\mathcal{D}(x) \in \{\text{Real}, \text{Fake}\}$ be the target detector. The attack receives only binary labels, without target scores, gradients, parameters, or intermediate features. It uses independently obtained labeled real and generated development images, a separate real-reference pool, and an optional fixed perceptual feature network. These auxiliary resources do not provide access to the target detector's training data or gradients. Their provenance and any overlap with detector training data must be documented.

A frozen perceptual distance d_p defines the quality constraint. For query budget $Q_{\max}$ and threshold δ_p, successful evasion requires

$$\mathcal{D}(x) = \text{Real}, \qquad d_p(x, x_o) \le \delta_p, \qquad x \in [0,1]^{H\times W\times 3}, \qquad q \le Q_{\max}. \tag{1}$$

The threshold and metric implementation are selected before test evaluation. Boundary navigation minimizes $\|x - x_o\|_2$; this is a proxy and does not guarantee a reduction in d_p. Every verified real-classified candidate is therefore also evaluated under the fixed quality constraint. An unrelated real reference is a feasible starting point for boundary navigation, not automatically a successful attack.

3.2 Spectral Artifact Profiling

For each RGB channel c, use a unitary two-dimensional discrete Fourier transform:

$$X_c(u,v) = \frac{1}{\sqrt{HW}} \sum_{m=0}^{H-1} \sum_{n=0}^{W-1} x_c(m,n) \exp\Big[-j2\pi \left(\frac{um}{H} + \frac{vn}{W}\right)\Big]. \tag{2}$$

The channel power spectrum is $P_{x,c} = |X_c|^2$. All spectra use the same image size, channel ordering, and transform normalization. From disjoint real and generated profiling sets, compute

$$\overline{P}_{r,c} = \frac{1}{N_r} \sum_i P_{x_{r,i},c}, \qquad \overline{P}_{f,c} = \frac{1}{N_f} \sum_i P_{x_{f,i},c}, \qquad \Delta P_c = \overline{P}_{r,c} - \overline{P}_{f,c}. \tag{3}$$

A positive discrepancy represents lower average generated-image power at that frequency; a negative discrepancy represents higher average power. These are distributional differences, not evidence of a detector's local sensitivity. Content, acquisition, or compression differences can also produce them.

3.3 Frequency Bands and Spectral Reconstruction

Construct radial bands using signed normalized FFT frequencies $f_H(u)$ and $f_W(v)$, with

$$\rho(u,v) = \sqrt{f_H(u)^2 + f_W(v)^2}, \qquad \Omega_b = \{(u,v) : r_{b-1} \le \rho(u,v) < r_b\}. \tag{4}$$

The final band includes its outer endpoint. Empty bands are omitted. DC is excluded from correction and sampling to avoid explicitly changing channel means; it is restored exactly before inversion. The boundaries are fixed before testing. FFT shifting is used only for visualization; the computational arrays remain in the unshifted convention.

Write $X_{o,c} = A_{o,c} e^{j\phi_{o,c}}$. One nonnegative coefficient α_b is shared by the three channels within band b. For non-DC frequencies,

$$\widetilde{P}_c(u,v;\boldsymbol{\alpha}) = \max\{P_{o,c}(u,v) + \alpha_{b(u,v)} \Delta P_c(u,v), \epsilon\}, \qquad \epsilon > 0. \tag{5}$$

Reconstruct

$$T(x_o;\boldsymbol{\alpha}) = \mathrm{clip}_{[0,1]} \left(\mathrm{Re}\, \mathcal{F}^{-1} \left[\sqrt{\widetilde{P}_c}\, e^{j\phi_{o,c}} \right]_{c=1}^{3} \right). \tag{6}$$

At DC, the original complex coefficient replaces the corrected one. Symmetric radial masks and real-image power spectra preserve conjugate symmetry; implementations explicitly pair coefficients to control numerical error. Self-conjugate coefficients remain real. Phase at zero-magnitude coefficients is assigned consistently, but carries no original structural information. Phase is retained before pixel clipping, which can change the final spectrum. The method does not guarantee imperceptibility or preservation of all semantic details.

3.4 Adaptive Spectral Alignment

Within each band and channel, normalize the corrected and reference power:

$$q_{c,b}(u,v;\boldsymbol{\alpha}) = \frac{\widetilde{P}_c(u,v;\boldsymbol{\alpha})}{\sum_{(m,n)\in\Omega_b} \widetilde{P}_c(m,n;\boldsymbol{\alpha})}, \qquad q_{r,c,b}(u,v) = \frac{\overline{P}_{r,c}(u,v) + \epsilon}{\sum_{(m,n)\in\Omega_b} (\overline{P}_{r,c}(m,n) + \epsilon)}. \tag{7}$$

ASA follows the updated methodology by balancing spectral alignment and a full-image perceptual penalty:

$$J(\boldsymbol{\alpha}) = \frac{1}{3B} \sum_{c=1}^{3} \sum_{b=1}^{B} D_{\mathrm{KL}}(q_{c,b}(\boldsymbol{\alpha}) \| q_{r,c,b}) + \lambda_{\mathrm{perc}} d_p(T(x_o;\boldsymbol{\alpha}), x_o), \qquad \boldsymbol{\alpha} \in [0, \alpha_{\max}]^B. \tag{8}$$

Here $D_{\mathrm{KL}}(q\|p) = \sum q \log(q/p)$ and $\lambda_{\mathrm{perc}} \geq 0$. The perceptual network, if used, is independent of the detector. It supplies a local image-quality objective, not target feedback.

Because reconstruction couples bands through the perceptual term, coefficients are optimized by coordinate descent rather than independent one-dimensional minimizations. Initialize $\boldsymbol{\alpha} = \mathbf{0}$, evaluate a fixed finite coefficient grid for each band while holding the others fixed, and retain the lowest full-objective value. Sweep bands in a fixed order until the relative objective decrease is below a prescribed tolerance or the sweep limit is reached. Grid ties select the smaller coefficient. Record the grid, maximum coefficient, sweep limit, tolerance, and numerical epsilon. This procedure produces an approximate local solution; no unimodality or global-optimum guarantee is assumed. Set $x_{\mathrm{spec}} = T(x_o; \boldsymbol{\alpha}^*)$.

Band normalization aligns spectral shape while discarding total band energy. Therefore Equation 8 does not guarantee energy matching. An energy-aware objective is deferred to future work to keep the present evaluation focused on the updated method. Report pre- and post-clipping spectral discrepancies, clipping fraction, and image distortion as diagnostics from the retained runs.

3.5 Initialization and Query Accounting

ASA construction uses no target queries. One detector call verifies x_{spec}. If Equation 1 holds, return it. If it is real-classified but exceeds the quality threshold, retain it as a boundary-refinement starting point rather than terminating.

If x_{spec} is fake-classified, select references in a deterministic, predeclared order from a held-out real pool. Count each verification query. For the first real-classified reference, bracket the boundary on the segment

$$x(\beta) = (1 - \beta)x_o + \beta x_{\mathrm{ref}}, \qquad \beta \in [0, 1]. \tag{9}$$

Maintain one fake endpoint and one real endpoint; query their midpoint and replace the endpoint with the same label. Return the real endpoint at the fixed tolerance, iteration cap, or query limit. This locates a label transition in the bracket without assuming that the full segment has a single transition. Using x_o, rather than x_{spec}, makes the fallback initialization consistent with the updated method and the distortion reference. Consequently, in ASA-failure cases, ASA influences SBQR through the weights but does not directly determine the interpolation endpoint. If no real-classified point is found within budget, record an initialization failure.

A single counter includes ASA verification, reference selection, binary search, and refinement. Offline profiling and local perceptual evaluations consume computation but no target queries. Initial clean-image screening is a separately reported eligibility cost, excluded consistently from attack budgets for every method. Within a run, previously queried labels are cached; any additional target call is counted. A detector call is one complete label inference, including the target's internal reconstruction pipeline.

3.6 Spectral-Biased Query Refinement

For non-DC frequencies, define

$$a(u, v) = \frac{1}{3} \sum_c |\alpha^*_{b(u,v)} \Delta P_c(u, v)|. \tag{10}$$

Let M be the number of non-DC frequency coordinates. Use an explicit exploration mixture:

$$W(u, v) = (1 - \zeta) \frac{a(u, v)}{\sum a} + \frac{\zeta}{M}, \qquad 0 < \zeta \leq 1, \tag{11}$$

with $W = 1/M$ when $\sum a = 0$, and $W(0, 0) = 0$. This replaces a scale-dependent additive floor with a documented exploration fraction. It allocates expected perturbation energy, not coordinate-selection probabilities, and retains support over the entire non-DC space.

For each channel sample a conjugate-symmetric complex Gaussian spectrum $Z_{t,c}$, with unit expected squared magnitude at ordinary frequency pairs and real Gaussian self-conjugate coefficients. Set DC to zero and generate

$$s_{t,c} = \mathrm{Re}\, \mathcal{F}^{-1}(\sqrt{W} \odot Z_{t,c}). \tag{12}$$

Let x_t be the current real-classified point and $d_t = x_o - x_t$. Project and normalize across all pixels and channels:

$$\widehat{\eta}_t = s_t - \frac{\langle s_t, d_t \rangle}{\|d_t\|_2^2} d_t, \qquad \eta_t = \widehat{\eta}_t / \|\widehat{\eta}_t\|_2. \tag{13}$$

Terminate if $\|d_t\|_2$ is below numerical tolerance; resample a numerically degenerate projected direction within a fixed local cap. Projection modifies the original spectral allocation, so the post-projection spectrum need not equal W.

Following the updated boundary procedure, form

$$x_t^{\perp} = \mathrm{clip}_{[0,1]}(x_t + \gamma_t \|d_t\|_2 \eta_t), \qquad x_t' = \mathrm{clip}_{[0,1]}[x_t^{\perp} + \lambda_t(x_o - x_t^{\perp})]. \tag{14}$$

Before querying, reject candidates that do not reduce $\|x_t' - x_o\|_2$ by the prescribed numerical tolerance. Otherwise query once and accept only a real prediction. Without clipping, contraction requires $(1 - \lambda_t)\sqrt{1 + \gamma_t^2} < 1$; the explicit distance check also handles clipping. Accepted updates decrease Euclidean distance, but not necessarily perceptual distance.

Initialize (γ_0, λ_0) to the prespecified values in the protocol. Over each fixed-length proposal window, include both local distance rejections and detector rejections in the acceptance fraction. If this fraction is below p_{low}, divide both step sizes by $a_{\mathrm{step}} > 1$; if above p_{high}, multiply both by a_{step}; otherwise retain them. Clip step sizes to predeclared bounds, with $0 < \lambda_t < 1$. Freeze the window length, thresholds, bounds, stagnation tolerance, proposal cap, and adaptation factor before testing. The proposal cap prevents unlimited local work when no detector queries are consumed.

Maintain an archive of queried real-classified candidates. At each query checkpoint report the archive point satisfying $d_p \leq \delta_p$ with smallest Euclidean distortion; if none exists, report failure together with the best real-classified diagnostic candidate. Stop on first quality-constrained success for query-to-success experiments. For fixed-budget quality comparisons, continue refinement and retain the best feasible archived point. All baselines use the corresponding stopping mode.

3.7 Complete Procedure

1. Load the frozen SAP prior and eligible source; initialize the query counter and candidate archive.
2. Optimize Equation 8, reconstruct the ASA candidate, and compute spectral weights.
3. Query the ASA candidate once if budget remains. Archive it if real-classified; return only if the quality constraint and stopping mode permit.
4. If ASA is fake-classified, verify references and bracket on the source–reference segment, checking remaining budget before every call. Record failure if no starting point is obtained.
5. Sample, project, and construct proposals. Reject non-improving candidates locally. Query improving candidates once; archive and accept only real-classified points. Apply the frozen adaptation rule.
6. Stop at the query, proposal, or stagnation limit, or quality-constrained success in first-success mode. Return the best feasible archived point, or explicit failure with a diagnostic output.

4 Experimental Protocol and Pending Results

No execution logs or measured experimental outputs accompany this revision. Accordingly, the following is a proposed protocol. Dataset counts describe the intended selection and must be verified against an image manifest. There are no reported ASR values, statistical findings, or completed result figures.

4.1 Data and Separation

The planned selection from the benchmark associated with DIRE [3] contains 1,000 real images, 500 Stable Diffusion v1 images, and 500 Stable Diffusion v2 images. The proposed partition is shown in Table 1. Only generated test images are attack targets. The real test subset assesses clean false-positive rates; it is not the reference pool. Reserve the reference pool from real development images and record its identities separately. Profiling may use those real development images, but their reuse for initialization is disclosed. Record source URLs, dataset version, generator identifiers, file hashes, split seed, duplicate checks, and prompt or source-image grouping where available. Apply identical preprocessing within the evaluation interface, while acknowledging that it cannot remove pre-existing acquisition or compression differences. Check overlap with target-detector training data when metadata permits, and state unresolved overlap explicitly.

Table 1: Proposed partition, pending manifest verification.

Subset	Real	SD-v1	SD-v2	Total
Development	500	250	250	1000
Validation	100	50	50	200
Test	400	200	200	800

For cross-generator evaluation, construct SAP from one generator's development images and evaluate only the other generator's held-out images. Use source-only generated validation images for attack hyperparameters in a strict transfer setting; any target-generator tuning is reported as an adapted setting. Keep the real development subset fixed. A mixed-generator prior is an in-domain reference, not evidence of unseen-generator generalization. SD-v1–SD-v2 transfer alone supports only a narrow generalization claim.

4.2 Detectors and Baselines

Use two primary detector families: UniversalFakeDetect and DIRE [2, 3]. Record checkpoint hashes, implementation commits, input transforms, binary thresholds, and inference settings. Freeze thresholds without attack-test tuning and freeze reconstruction randomness for the deterministic primary evaluation. Each query must recompute the detector representation from the attacked RGB image; in particular, cached clean DIRE representations cannot substitute for attacked-image reconstruction.

Compare complete pipelines for SAGE, BA, HSJA, and QEBA [9, 10, 11]. Retain one documented spatial QEBA variant with its implementation defaults recorded before test access. All methods receive the same real-reference pool and comparable validation access. Count every online initialization and refinement call. Report SAGE's auxiliary profiling and perceptual-network requirements and local computation separately. The scope is comparison with representative label-only attacks, not exhaustive state-of-the-art superiority.

4.3 Compact Evaluation Design

The following three blocks constitute the proposed compact evaluation. Their size is an author-selected evidence plan, not a journal-prescribed minimum or an acceptance guarantee.

1. **Primary comparison.** Evaluate the four complete pipelines on both detectors and all 400 generated test images (200 per generator), subject to manifest verification and clean eligibility. Use three predeclared seeds, proposed as 0, 1, and 2. Run each attack once per seed with a maximum of 5,000 queries, recording checkpoints at 500, 1,000, 2,000, and 5,000. These checkpoints are extracted from the same trajectory, not independent experiments. Stop on the first quality-constrained success and carry its success indicator forward to later checkpoints. Report generator-specific and pooled results.

2. **Focused component controls.** Use a stratified, manifest-fixed subset of 100 generated test images (50 per generator), selected before observing attack outcomes, on both detectors with the same seeds. Obtain one common verified starting point per image and seed using reference initialization; charge its full cost to each controlled variant. Compare the identical refinement engine with (a) uniform weights, (b) fixed low-frequency weights, and (c) ASA-derived learned weights. For (b), use Equation 11 with $a(u, v) = \mathbb{I}\{0 < \rho(u, v) \leq 0.25\}$ and the same exploration fraction ζ; this is a control, not a reproduction of QEBA. Compute ASA coefficients for (c), but bypass the ASA starting-point shortcut. Compare the complete SAGE pipeline against this controlled learned-weight variant on the identical subset, reusing the primary runs, to assess the operational contribution of ASA candidate initialization. ASA's one-query constrained success rate is also obtained directly from primary logs. This design distinguishes initialization from sampling while avoiding a larger factorial grid. Initialization failure is reported for all controlled variants rather than dropping the image.

3. **Narrow source-only transfer.** Select SD-v1 as the source and SD-v2 as the held-out target before execution. Build SAP using only SD-v1 development images and retain the prespecified attack settings and use only source-generated validation images for implementation checks. Evaluate SAGE on the 200 SD-v2 test images against both detectors with the same seeds. Reuse the primary

baseline runs; compare the source-only and mixed-prior SAGE results as distinct settings. No reverse transfer or target tuning is included. These outcomes support only transfer between these two generator versions.

The primary comparison entails at most 9,600 runs; the three controlled variants add at most 1,800 runs; source-only transfer adds at most 1,200 runs. Thus the upper bound is 12,600 runs and 63 million attack-stage label queries, before clean screening and validation. Eligibility and early stopping reduce this bound. DIRE reconstruction can make even this compact design expensive; benchmark wall time on validation images before committing resources.

Retain two main result tables (pipeline performance and component/transfer differences), one ASR-versus-query figure, and one representative success/failure figure. Clean detector performance, clipping statistics, timing, and image-quality summaries come from the same runs. Defer a third detector, additional transformation baselines, reverse transfer, shuffled priors, fixed-coefficient correction, energy-aware objectives, broad parameter sweeps, and separate compression/blur robustness experiments. These omissions narrow the supported claims. Investigate targeted diagnostics only if the retained runs reveal an unresolved failure or confound.

4.4 Prespecified Configuration and Execution Lock

Protocol SAGE-v1.0 fixes the design choices below before test access. These are prespecified defaults, not settings selected by completed validation or proven optimal. The accompanying JSON is the authoritative numerical configuration.

- Use a 256×256 RGB attack canvas, bicubic resizing, float32 values in $[0, 1]$, and no lossy roundtrip. Preserve official deterministic detector adapters and record their exact transforms. LPIPS uses the calibrated AlexNet version 0.1 metric in evaluation mode on $2x - 1$, with $\delta_p = 0.05$. This is an operational threshold, not a validated universal imperceptibility limit.

- Use eight equal-width radial bands over $[0, \sqrt{0.5}]$, unitary FFTs, separate RGB power spectra, excluded DC, and $\epsilon = 10^{-12}$. ASA uses $\alpha_b \in \{0, 0.1, \ldots, 1.0\}$, zero initialization, low-to-high coordinate sweeps, at most three sweeps, relative improvement tolerance 10^{-4}, and $\lambda_{\text{perc}} = 1$. Break numerical ties toward smaller coefficients; no energy penalty is added.

- Set $\zeta = 0.1$, $\gamma_0 = \lambda_0 = 0.01$, and bounds $[10^{-5}, 0.1]$ for both steps. Adapt every 50 proposals with acceptance thresholds 0.1/0.3 and factor 1.5. Include local rejections. Require a relative Euclidean improvement greater than 10^{-8} before querying; cap local proposals at 100,000 and degenerate-direction resampling at ten. Disable a separate stagnation stop.

- Check at most 50 references in one seeded order shared by methods. Boundary bracketing uses a coefficient tolerance 10^{-5} and at most 20 midpoint queries. Charge reference verification, ASA verification, and binary search to the 5,000-query budget. Use seeds 0/1/2 and checkpoints 500/1,000/2,000/5,000. Stop at first constrained success; quality summaries concern that stopping mode, not the minimum distortion at the full budget.

- Use split seed 20261008, ablation-selection seed 20261009, and bootstrap seed 20261010. Remove or group duplicates and known related prompt/source images before partitioning. Use 10,000 generator-stratified paired image bootstrap replicates for 95% intervals, retaining all seeds within images.

A validation-only pilot uses 40 generated validation images (20 per generator) to check implementation, eligibility, and runtime. It has not been executed. Protocol v1.0 does not tune these values from pilot ASR. If a correction or redesign is necessary, increment the version and freeze it before accessing test outcomes.

The design is fixed, but an execution lock remains incomplete because no data manifest, downloaded checkpoint, installed baseline implementation, or verified detector adapter accompanies this revision. Before execution, record their hashes, commits, full baseline parameter dictionaries, official decision thresholds, DIRE reconstruction settings, software versions, and hardware. Use one recorded default implementation configuration for each baseline, including a spatial QEBA variant, under a shared query wrapper. Do not invent resource identifiers or silently substitute models. Verify all target inference steps on attacked RGB candidates. These resource fields are deployment requirements rather than additional experimental sweeps.

4.5 Evaluation Metrics

For detector $\mathcal{D}$, use the eligible set

$$\mathcal{I}_{\mathcal{D}} = \{i : \mathcal{D}(x_{o,i}) = \text{Fake}\}. \tag{15}$$

Report its size and the clean fake-detection rate, together with the real-image false-positive rate. All methods use the same eligible images for a given detector. Across detectors, also report a common eligible subset if direct paired comparisons are made.

Let $\mathcal{V}_i(Q)$ contain all real-classified candidates actually verified within Q attack-stage queries, and define

$$S_i(Q, \delta_p) = \mathbb{I}\{\exists x \in \mathcal{V}_i(Q) : d_p(x, x_{o,i}) \le \delta_p\}, \qquad \text{ASR@}Q, \delta_p = \frac{\sum_{i \in \mathcal{I}_{\mathcal{D}}} S_i(Q, \delta_p)}{|\mathcal{I}_{\mathcal{D}}|}. \tag{16}$$

The first-success query is

$$Q_{\text{succ},i} = \min\{q : S_i(q, \delta_p) = 1\}. \tag{17}$$

Unsuccessful attacks are right-censored at the budget; they are not silently discarded. Present ASR-versus-query curves, failure and initialization-failure rates, and queries on a declared common-success subset. If reporting a budget-penalized mean, define it explicitly as replacing failures by $Q_{\max} + 1$ rather than treating that penalty as an observed successful query count. A median among successful runs is a conditional diagnostic, not the sole efficiency measure. Relative query ratios use the same eligible sample and the same failure treatment.

ASA's direct constrained evasion rate is

$$\text{ASA-SR} = \frac{1}{|\mathcal{I}_{\mathcal{D}}|} \sum_{i \in \mathcal{I}_{\mathcal{D}}} \mathbb{I}\{\mathcal{D}(x_{\text{spec},i}) = \text{Real},\ d_p(x_{\text{spec},i}, x_{o,i}) \le \delta_p\}. \tag{18}$$

ASA generation uses zero detector feedback, but success verification costs one query. Also report its label-only flip rate and distortion distribution separately; avoid calling the full attack literally query-free.

Retain LPIPS, PSNR, and root-mean-square pixel distortion using documented implementations and data ranges. LPIPS supplies the quality constraint; PSNR and pixel distortion are diagnostics derived from the same images. Verify and add the LPIPS metric reference before submission. Report quality on all attempted images with explicit failure handling, plus paired common-success comparisons and quality-constrained ASR. Include representative successes and failures, amplified residuals, and clipping diagnostics. Perceptual metrics do not by themselves establish semantic equivalence or universal human imperceptibility. Separate offline profiling time, local per-image optimization time, and complete detector-inference time.

5 Results

Table 2: Primary result

Method	ASR@1k	ASR@5k	Penalized queries	LPIPS*
BA	-	-	-	-
HSJA	-	-	-	-
QEBA	-	-	-	-
SAGE	-	-	-	-

6 Limitations and Threats to Validity

The prior is a mean spectral difference estimated from unpaired images; it may encode scene, brightness, compression, or acquisition differences rather than generation artifacts. Matching normalized within-band spectra does not guarantee matching band energy or detector evidence. Nonlinear detectors can depend on local or reconstruction cues not captured by global PSD, and zero-mean spectral sampling does not directly follow the signed correction direction.

ASA and spectral-weight construction require auxiliary real and generated images. Their offline cost, dependence on generator availability, and provenance limit the scope of a label-only claim. The current

Table 3: Component controls and source-only transfer.

Variant	ASR@1k (%)	ASR@5k (%)	Penalized queries
Panel A: Component controls			
Shared start + uniform sampling	-	-	-
Shared start + low-frequency sampling	-	-	-
Shared start + learned sampling	-	-	-
Complete SAGE	-	-	-
Panel B: SD-v1-to-SD-v2 transfer			
BA	-	-	-
HSJA	-	-	-
QEBA	-	-	-
SAGE: mixed-generator prior	-	-	-
SAGE: SD-v1-only prior	-	-	-

Report a separate table for each detector. Panel A uses the predefined 100-image subset; complete SAGE results are reused from the main experiment. Panel B uses SD-v2 test images; baseline and mixed-prior results are reused from the main experiment. ASR requires a Real prediction and LPIPS ≤ 0.05. Penalized queries assign 5,001 to unsuccessful runs. Report eligible image counts and image-level bootstrap 95% confidence intervals with the measured results.

proposed benchmark and two closely related generator versions offer limited coverage. A fixed local detector interface also differs from an API with stochastic preprocessing, rate limits, quantization, or changing decision thresholds. Such deployment conditions are not established by the proposed primary experiment.

Phase retention holds before clipping, and the perceptual penalty is a surrogate. Euclidean contraction during SBQR need not improve perceptual quality. Reference initialization may preserve substantial unrelated content if the budget is insufficient; these outcomes fail the quality-constrained criterion. The attack evaluation measures evasion of image detectors and does not establish that an output is genuinely photographic or that every forensic system is vulnerable.

7 Conclusion

SAGE proposes a shared spectral prior for detector-independent image correction and decision-based refinement. The updated method balances normalized spectral alignment with full-image perceptual distortion, uses a verified boundary starting point when necessary, and counts all initialization and refinement queries. Its evaluation requires both a real prediction and a fixed quality constraint. The proposed experiments will determine whether this design provides a useful efficiency–quality trade-off. Empirical conclusions remain pending.

Declarations to Finalize Before Submission

Author contributions: [To finalize: Confirm actual contributions with all authors.]
Funding: [To finalize: Confirm funding sources or absence of funding.]
Data and code availability: [To finalize: Add verified dataset provenance, split manifest, code repository and release identifier; describe access restrictions.]
Conflicts of interest: [To finalize: Obtain author confirmation.]
AI-assisted preparation: [To finalize: Document the tools and actual uses, author verification, and the journal's applicable disclosure wording.]
Ethics and consent applicability: [To finalize: Determine applicability from the actual datasets and journal requirements.]

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